



## C3 Photosynthesis

### C · Interaction and interdependence · IB Diploma · Concept guide and original exam-style practice

#### Understand the concept

Photosynthesis converts light energy into chemical energy stored in organic molecules. In the light-dependent reactions (thylakoid membrane), photosystems absorb light and use the energy to split water (photolysis), releasing oxygen, and to generate ATP (via chemiosmosis) and NADPH. In the light-independent reactions (Calvin cycle, stroma), the enzyme rubisco fixes CO<sub>2</sub> onto RuBP, and ATP and NADPH from the light-dependent reactions are used to reduce the fixed carbon to G3P, which regenerates RuBP and can be used to build glucose. The overall rate of photosynthesis is limited by whichever factor — light intensity, CO<sub>2</sub> concentration, or temperature — is furthest from optimal at a given time.

#### Worked example: follow the reasoning

A student measures the rate of photosynthesis of an aquatic plant at increasing light intensity, first at a low CO<sub>2</sub> concentration (0.02%) and then at a higher CO<sub>2</sub> concentration (0.1%), keeping temperature constant. At low CO<sub>2</sub>, rate rises with light intensity but plateaus early at a low maximum rate. At high CO<sub>2</sub>, rate rises further before plateauing at a higher maximum rate. Step 1: at low light intensity in both curves, light is limiting, since rate increases as light increases and both curves overlap. Step 2: at higher light intensity, the low-CO<sub>2</sub> curve plateaus while the high-CO<sub>2</sub> curve continues rising, showing CO<sub>2</sub> was limiting the low-CO<sub>2</sub> curve at that point (not light). Step 3: eventually the high-CO<sub>2</sub> curve also plateaus at a higher rate, at which point another factor (e.g. temperature or enzyme/rubisco capacity) becomes limiting. Conclusion: whichever factor is in shortest supply relative to the others at a given light intensity limits the overall rate of photosynthesis.

#### Quick check · multiple choice

In the light-dependent reactions, what immediately replaces the electrons lost from photosystem II?

- A Electrons from NADPH
- B Electrons from the splitting of water (photolysis)
- C Electrons generated in the Calvin cycle
- D Electrons released from CO<sub>2</sub> fixation

Write your choice: \_\_\_\_\_

#### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student investigated the effect of CO<sub>2</sub> concentration on the rate of photosynthesis of a plant at high, constant light intensity and constant temperature (25°C). Rate increased steadily as CO<sub>2</sub> concentration rose from 0.01% to 0.06%, then remained constant as CO<sub>2</sub> concentration was increased further to 0.10%. [6]
  - (a) Identify the limiting factor for photosynthesis between 0.01% and 0.06% CO<sub>2</sub>, and explain your reasoning. [2]
  - (b) Suggest, with a reason, what factor is most likely limiting the rate above 0.06% CO<sub>2</sub>. [2]
  - (c) Explain, in terms of the Calvin cycle enzymes, why the rate cannot increase indefinitely simply by adding more CO<sub>2</sub>. [2]

Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [6]

- (a) CO<sub>2</sub> concentration is the limiting factor between 0.01% and 0.06%, because increasing CO<sub>2</sub> while light and temperature remain constant directly increases the rate of photosynthesis, showing CO<sub>2</sub> availability was restricting the rate.
- (b) Above 0.06%, another factor — most likely light intensity or temperature (both held constant in this experiment) — becomes limiting, since increasing CO<sub>2</sub> further no longer increases rate.
- (c) CO<sub>2</sub> is fixed onto RuBP by the enzyme rubisco in the Calvin cycle; once all rubisco (and other Calvin cycle enzyme) molecules are working at their maximum rate, they become saturated and cannot process CO<sub>2</sub> any faster, so adding more CO<sub>2</sub> beyond this point cannot increase the rate of carbon fixation.

### Quick check answer

B. Photolysis splits water into electrons (replacing those lost from photosystem II), protons, and oxygen gas, which is released as a by-product.

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.